

Zephylin Dusengimana

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RESEARCH INTERESTS

- NLP, LLMs, Trustworthy AI, AI applications in High-Stakes Settings (Education, Healthcare).

EDUCATION

Oklahoma Christian University <i>Master of Science in Computer Science, Artificial Intelligence</i> GPA: 4.0/4.0	Oklahoma, USA Aug. 2024 – Dec. 2025
Kyungdong University <i>Bachelor of Science in Computer Engineering</i> GPA: 4.38/4.5 (WES: 4.0/4.0) Top Student	Goseong, South Korea Sep. 2020 – Aug. 2024

RESEARCH EXPERIENCE

Undergraduate Research Assistant <i>AI and Computer Vision Lab</i>	Mar. 2022 – May 2024 Goseong, South Korea
- Designed and evaluated deep learning models with systematic ablation studies, measuring generalization and robustness across varied real-world conditions - Built reproducible data preprocessing and augmentation pipelines for heterogeneous datasets, supporting rigorous model comparison and error analysis - Quantified a 23-point domain gap in detection accuracy between close-range and distant viewing conditions, and validated a mixed-scale training strategy that closed most of the gap	

PROFESSIONAL EXPERIENCE

AI Software Engineer <i>AfiaSync</i>	Jan. 2026 – Present Atlanta, GA
- Designed a Python LLM-as-a-judge evaluation pipeline benchmarking model outputs across 30 scenarios, scoring quality against a standardized rubric and cutting iteration cycles from hours to under 60 seconds - Engineered a versioned prompt framework with JSON schema enforcement and hallucination guardrails, increasing rubric-based quality scores from 40% to 80% - Built an end-to-end NLP pipeline over speech and text using GPT-4o, transforming unstructured input into structured, evaluated outputs	
AI Software Engineer Intern <i>AfiaSync</i>	June 2025 – Dec. 2025 Remote
- Implemented an LLM-powered assistant using retrieval-augmented generation (RAG) and embedding-based semantic search to improve answer relevance and grounding - Developed retrieval and reranking components over a knowledge base, improving factual consistency of generated responses	
Machine Learning Engineer Intern <i>HighView Tech Group</i>	Jan. 2024 – Apr. 2024 Kigali, Rwanda
- Built anomaly detection and predictive modeling pipelines over large-scale sensor data using Python, pandas, and NumPy, improving data quality and reliability - Collaborated with engineers to integrate models into production data systems with reproducible evaluation	

SELECTED PROJECTS

AI Resume Optimization Agent <i>Llama 3.1, Ollama, NLP, Prompt Engineering, LaTeX, Python</i>	
- Developed an NLP system that analyzes documents against target descriptions and generates tailored, structured outputs, using semantic matching and content ranking across multiple source versions - Designed automated evaluation and validation logic to preserve factual consistency and output quality, reducing manual effort by 85% - Applied prompt engineering and structured generation pipelines with local open-source LLMs for reproducible, low-cost inference	

Multimodal RAG & Retrieval System | *Python, Hugging Face, LangChain, FAISS, PyTorch*

- Built a retrieval-augmented system integrating transformer models with dense-embedding semantic search over heterogeneous documents (text, PDFs, images)
- Engineered retrieval and reranking pipelines and an evaluation framework measuring retrieval precision and answer reliability across varied queries

Agentic AI Shopping Assistant – Vesoko | *Python, LangGraph, MCP, FastAPI, GPT-4o*

- Architected a 15+ tool MCP server and LangGraph ReAct agent for query understanding, retrieval, and multi-step decision support in a production platform
- Built an LLM pipeline for multilingual text generation with automated quality insights and optimization recommendations

TECHNICAL SKILLS

Languages: Python, C++, SQL
NLP & LLMs: Transformers, Hugging Face, LangChain, LangGraph, NLTK, RAG, Semantic Search, LLM Evaluation, MCP, AI Agents
Machine Learning: PyTorch, TensorFlow, Scikit-learn, Embeddings, FAISS, Data Augmentation, Anomaly Detection
Data Analytics: Pandas, NumPy, Matplotlib, Seaborn, ETL/ELT Pipelines
Tools & DevOps: Git, Docker, CI/CD, GitHub Actions, Azure, AWS, Jupyter, Google Colab

HONORS & AWARDS

- **Top Student**, Computer Engineering, Kyungdong University 2024
- **3× Winner**, KDU Hackathon Programming Contest, Kyungdong University 2021–2023
- **Winner**, RUDN International Math Olympiad, Russia 2020
- **Winner**, Rwanda National Mathematics Competition 2020